Data Cleaning Report

Project: Employee Attrition Analysis

1. Dataset Overview

This report documents the data preprocessing and quality assurance steps taken to prepare the raw IBM HR Analytics dataset (`WA_Fn-UseC_-HR-Employee-Attrition.csv`) for exploratory data analysis and dashboard visualization. The goal of this phase was to ensure data integrity, remove redundancies, and structure the dataset optimally.

Metric	Pre-Cleaning (Raw)	Post-Cleaning (Final)
Total Records (Rows)	1,470	1,470
Total Features (Columns)	35	31
Duplicate Rows	0	0
Missing Values (Nulls)	0	0

2. Data Quality Operations

2.1 Missing Values & Duplicates Verification

Initial programmatic checks using Pandas (`df.isnull().sum()` and `df.duplicated().sum()`) confirmed that the raw dataset was provided without any null values or duplicate records. No imputation or row-level deletion was required.

2.2 Feature Selection (Dimensionality Reduction)

During the data review, four features were identified as redundant or providing zero variance. These columns were permanently dropped from the working dataset to optimize processing and simplify the schema:

- **EmployeeCount:** Contains a static value of 1 for all records. Provides no analytical value.
- **StandardHours:** Contains a static value of 80 for all records. Provides no variance for comparison.
- **Over18:** Contains a static value of 'Y' for all records.
- **EmployeeNumber:** Serves merely as a unique identifier/index and holds no predictive value for attrition.

> **Dataset Integrity Verified**
>
> The resulting `cleaned_data.csv` contains 1,470 rows and 31 high-value features, strictly typed into continuous numerical and categorical classes, fully prepared for downstream Tableau ingestion.

3. Column Classification (Numerical vs Categorical)

After cleaning, the 31 remaining columns were grouped by data type so that each could be analysed appropriately. Rating-style columns (1 to 4 scales) are stored as numbers but are treated as ordered categories in the analysis, for example `JobSatisfaction` in the dashboard.

Class	Count	Columns
Categorical (text)	8	Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime
Ordinal-coded (rating scales)	9	Education, EnvironmentSatisfaction, JobInvolvement, JobLevel, JobSatisfaction, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, WorkLifeBalance
Continuous / count numerical	14	Age, DailyRate, DistanceFromHome, HourlyRate, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, TotalWorkingYears, TrainingTimesLastYear, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager

4. Post-Cleaning Validation Through the Dashboard

To confirm that the cleaned data was loaded correctly, `cleaned_data.csv` was connected to Tableau Public and the Employee Attrition Analysis dashboard was built on it. If the cleaning steps had removed or corrupted records, the dashboard totals would not match the cleaned file. They do match, as shown below.

4.1 Master Dashboard

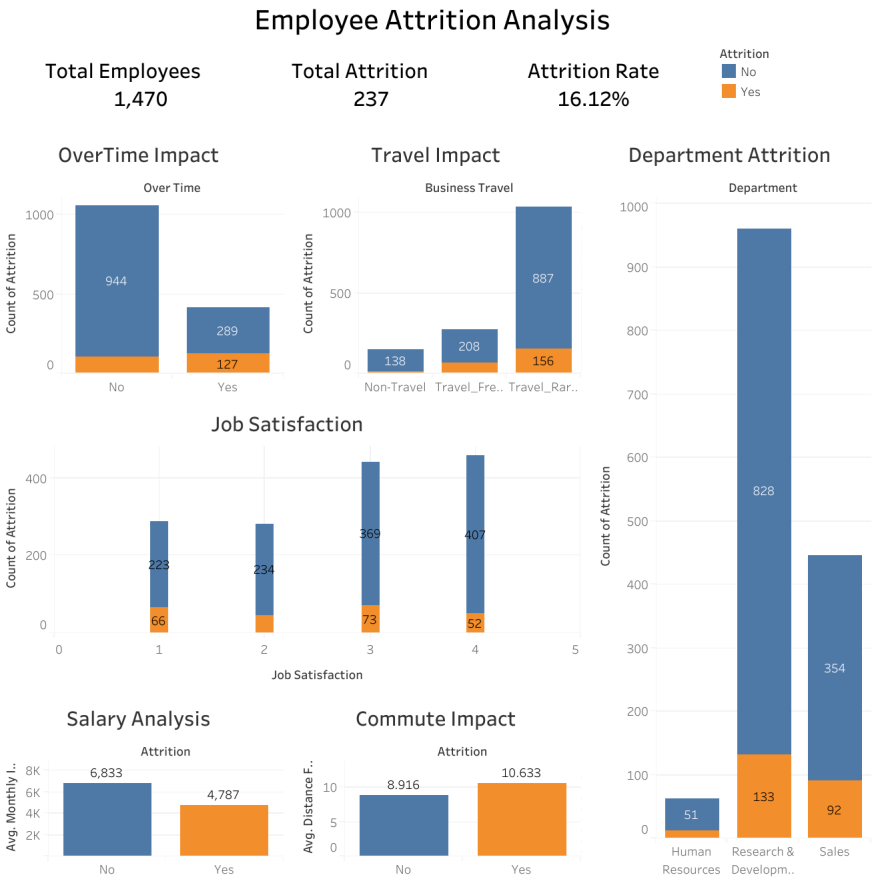


Figure 1: Employee Attrition Analysis dashboard built on cleaned_data.csv

Total Employees	Total Attrition	Attrition Rate
1,470	237	16.12%

Figure 2: Summary KPIs from the dashboard

The KPI strip reports **1,470 total employees**, **237 total attrition** and an **attrition rate of 16.12%** (retention 83.88%). The employee count equals the row count of the cleaned file, which confirms that no records were lost during cleaning, and the attrition rate is consistent with 237 / 1,470.

4.2 Record-Count Reconciliation

Every chart on the dashboard splits the same 1,470 employees into different segments. Adding up the retained employees (Attrition = No) in each chart should therefore always give 1,470 – 237 = 1,233. All four charts reconcile:

Chart	Retained employees by segment (Attrition = No)	Sum	Check
OverTime Impact	944 + 289	1,233	Matches
Travel Impact	138 + 208 + 887	1,233	Matches
Job Satisfaction	223 + 234 + 369 + 407	1,233	Matches
Department Attrition	51 + 828 + 354	1,233	Matches

The departed employees reconcile as well. In the OverTime chart, 110 + 127 = 237, which equals the total attrition on the KPI strip.

5. Visual Validation of Key Fields

Each graph below uses one or two cleaned columns. The charts confirm that these fields are populated, correctly typed and ready for analysis.

5.1 OverTime Impact

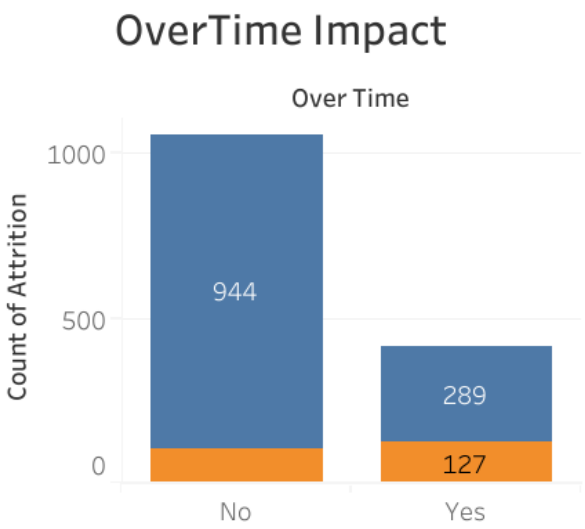


Figure 3: OverTime Impact

Fields used: `OverTime`, `Attrition` (both categorical).

What it shows: The number of employees who stayed (blue) and left (orange) for those who work overtime and those who do not.

Data check: Both fields have only two clean values (Yes/No) and no blanks. The four bars add up to all 1,470 employees.

5.2 Travel Impact

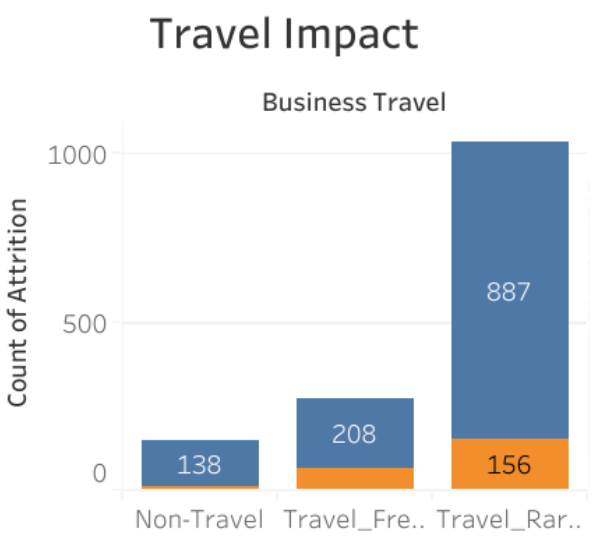


Figure 4: Travel Impact

Fields used: `BusinessTravel`, `Attrition`.

What it shows: Attrition across the three travel categories: Non-Travel, Travel_Frequently and Travel_Rarely.

Data check: The three categories are consistent with no spelling variants. Travel_Rarely is the largest group, with 156 employees who left.

5.3 Job Satisfaction



Figure 5: Job Satisfaction

Fields used: `JobSatisfaction` (ordinal 1 to 4), `Attrition`.

What it shows: Retained and departed employees for each satisfaction rating. Departures are 66, 73 and 52 for ratings 1, 3 and 4.

Data check: Only the valid ratings 1 to 4 appear, with no out-of-range values.

5.4 Salary Analysis

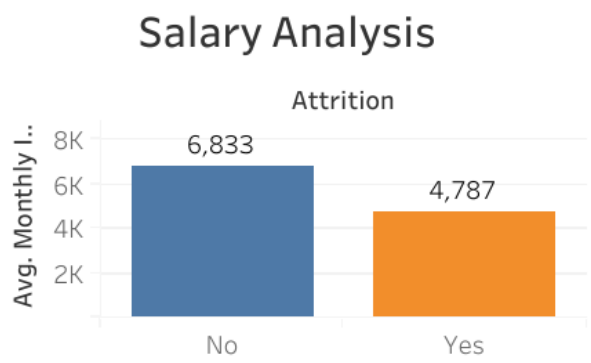


Figure 6: Salary Analysis

Fields used: `MonthlyIncome` (continuous), `Attrition`.

What it shows: The average monthly income of employees who stayed ($6,833) and those who left ($4,787).

Data check: Income is stored as a numeric value, so averages calculate correctly without any conversion.

5.5 Commute Impact

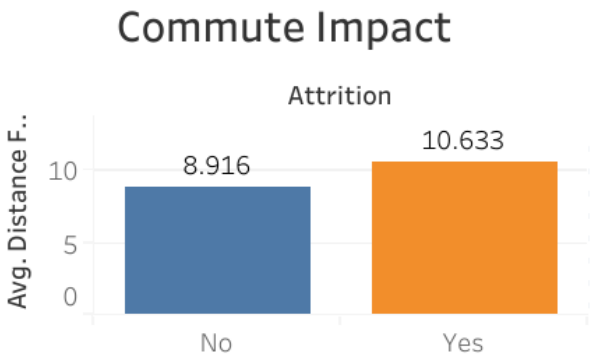


Figure 7: Commute Impact

Fields used: `DistanceFromHome` (continuous), `Attrition`.

What it shows: The average distance from home to office: 8.916 miles for employees who stayed and 10.633 miles for those who left.

Data check: Distance is numeric and complete, with no missing values to distort the averages.

5.6 Department Attrition

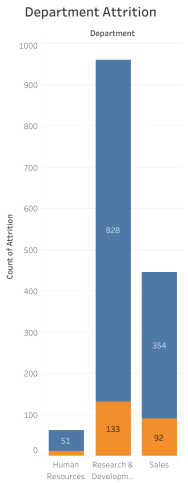


Figure 8: Department Attrition

Fields used: `Department`, `Attrition`.

What it shows: The count of retained and departed employees in Human Resources, Research & Development and Sales. R&D has the most departures (133), followed by Sales (92).

Data check: Only three department values appear. Retained employees add up to 51 + 828 + 354 = 1,233.

Conclusion

The cleaned dataset retained all 1,470 records and 31 usable features. Dashboard totals reconcile with the cleaned file, and every field used in the visualizations is complete and consistently typed. The data is ready for the exploratory analysis in the next report.